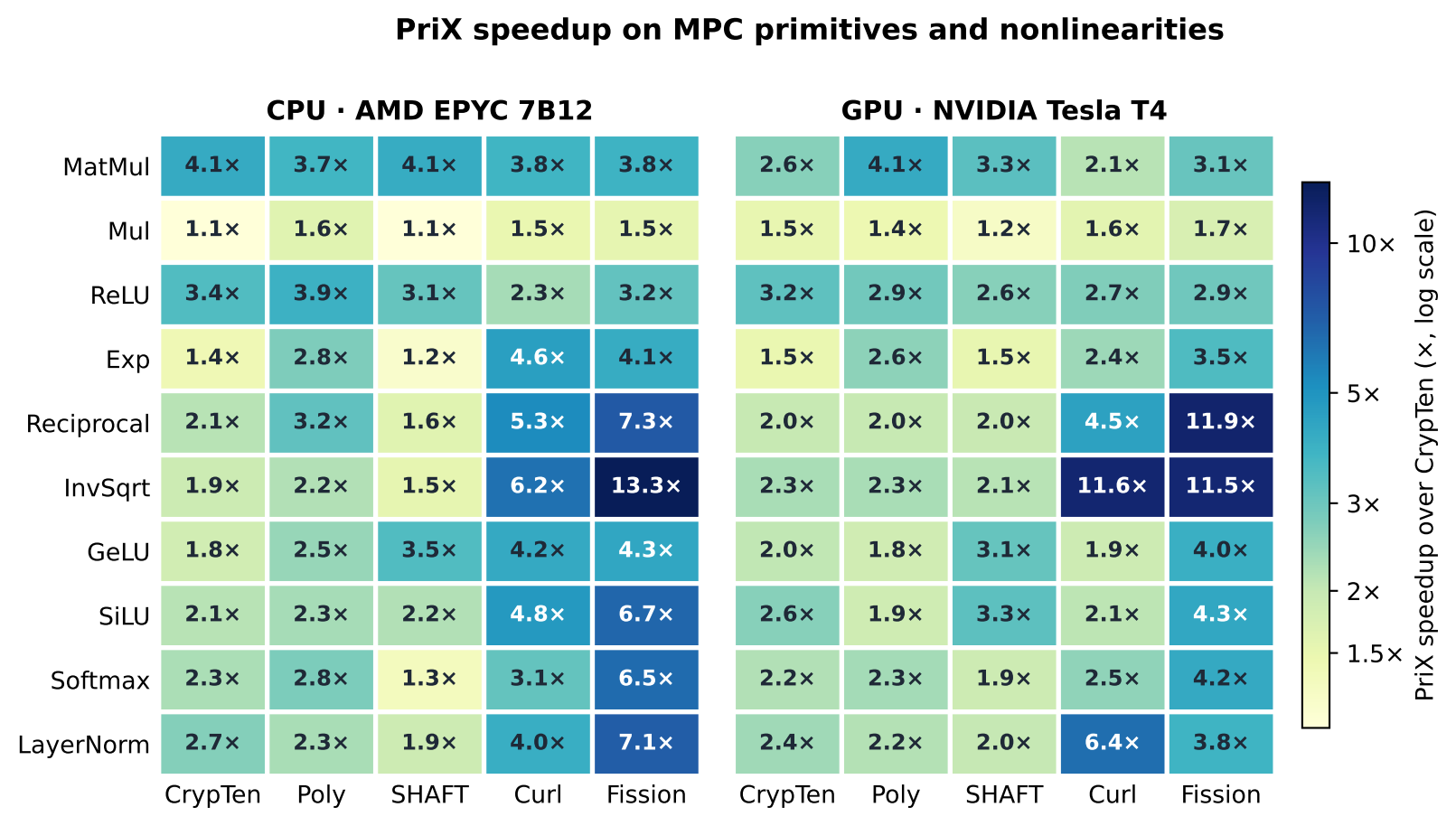
PriX speedup on MPC primitives and nonlinearities
CPU · AMD EPYC 7B12
GPU · NVIDIA Tesla T4
MatMul 4.1× 3.7× 4.1× 3.8× 3.8× 2.6× 4.1× 3.3× 2.1× 3.1×
Mul 1.1× 1.6× 1.1× 1.5× 1.5× 1.5× 1.4× 1.2× 1.6× 1.7×
ReLU 3.4× 3.9× 3.1× 2.3× 3.2× 3.2× 2.9× 2.6× 2.7× 2.9×
Exp 1.4× 2.8× 1.2× 4.6× 4.1× 1.5× 2.6× 1.5× 2.4× 3.5×
Reciprocal 2.1× 3.2× 1.6× 5.3× 7.3× 2.0× 2.0× 2.0× 4.5× 11.9×
InvSqrt 1.9× 2.2× 1.5× 6.2× 13.3× 2.3× 2.3× 2.1× 11.6× 11.5×
GeLU 1.8× 2.5× 3.5× 4.2× 4.3× 2.0× 1.8× 3.1× 1.9× 4.0×
SiLU 2.1× 2.3× 2.2× 4.8× 6.7× 2.6× 1.9× 3.3× 2.1× 4.3×
Softmax 2.3× 2.8× 1.3× 3.1× 6.5× 2.2× 2.3× 1.9× 2.5× 4.2×
LayerNorm 2.7× 2.3× 1.9× 4.0× 7.1× 2.4× 2.2× 2.0× 6.4× 3.8×
CrypTen Poly SHAFT Curl Fission CrypTen Poly SHAFT Curl Fission
10×
5×
3×
2×
1.5×
PriX speedup over CrypTen (×, log scale)